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Determinants of Return to Play After the Nonoperative Management of Hamstring Injuries in Athletes

A Systematic Review

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Background: It is important for clinicians to rely on suitable prognosis factors after hamstring injuries because of the high incidence of these injuries and time away from athletic activities.

Purpose: To summarize the current literature on factors that influence return to play after a hamstring injury in athletes.

Study Design: Systematic review.

Methods: A computer-assisted literature search of CINAHL, MEDLINE, Embase, and EBM Reviews databases (and a manual search of the reference lists of all selected articles) was conducted using keywords related to hamstring injuries and return to play. The literature review criteria included (1) patients with an acute hamstring or posterior thigh injury; (2) a randomized controlled trial, cohort study, case-control study, case series, or prospective or retrospective design; (3) information on rehabilitation, physical therapy, clinical assessment, imaging techniques, and return to play; and (4) studies written in English or French.

Results: The search strategy identified 914 potential articles, of which 24 met the inclusion criteria. In terms of the clinical assessment, the following factors were associated with a longer recovery time: stretching-type injuries, recreational-level sports, structural versus functional injuries, greater range of motion deficit with the hip flexed at 90°, time to first consultation >1 week, increased pain on the visual analog scale, and >1 day to be able to walk pain free after the injury. As for magnetic resonance imaging studies, the following factors correlated with a longer recovery time: positive findings; higher grade of injury; muscle involvement >75%; complete transection; retraction; central tendon disruption of the biceps femoris; proximal tendon involvement; shorter distance to the ischial tuberosity; length of the hamstring injury; and depth, volume, and large cross-sectional area. With respect to ultrasound studies, the following factors were associated with a poor prognosis: large cross-sectional area, injury outside the musculotendinous junction, hematoma, structural injury, and injury involving the biceps femoris. Lastly, rehabilitation approaches that included hamstring loading during extensive lengthening or 4 daily sessions of static hamstring stretching led to shorter rehabilitation times.

Conclusion: Numerous determinants have an effect on return to play after a hamstring injury in athletes. It is important for sports professionals to be aware of those determinants to guide athletes through the rehabilitation process and refine return-to-play strategies.

Keywords: hamstring injury; return to play; imaging; rehabilitation

Hamstring injuries are one of the most common soft tissue injuries affecting the lower extremities in athletes.[1,3,5,7-10,13,15,20,26] These injuries mostly occur in athletes involved in football, rugby, soccer, track and field, and dance.[6,7,10,11,13,20,26,28] They can be classified into 2 categories: a high-energy injury relating to a rapid change in muscle length in conjunction with a change in magnitude and rate of force development, and a lower energy stretching-type injury occurring at the extremes of muscle-lengthening positions.[1-4] The sprinting-type injury often occurs during the late swing through the midstance phase of running, whereas the stretching-type injury typically occurs in a position combining a large range of hip flexion and knee extension simultaneously. These injuries are responsible for a significant loss of time spent in competition.[1-3,6,8,9,11,17,26,28] Because of the high incidence of these injuries and the time spent away from competition, it is important for clinicians to rely on suitable prognostic factors.

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The purpose of this study was to conduct a review of the literature on the different factors that influence return to play after a hamstring injury in athletes. These determinants are typically available and obtained using clinical assessment, imaging, and rehabilitation techniques.

METHODS

Research Framework

The Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines were used during the search and reporting phase of this review.[14]

Eligibility Criteria

Articles were eligible if they met all of the following criteria: (1) patients were athletes with an acute hamstring injury; (2) a randomized controlled trial, cohort study, case-control study, case series, or prospective or retrospective design was used; (3) the study included clinical assessment, imaging technique, rehabilitation, physical therapy, and return-to-play information; and (4) the study was published in English or French. Articles were excluded (1) if the reported pathological abnormality was a contusion (created by an external force) or chronic tendinopathy or (2) if the treatment consisted of surgery.

Information Sources

A systematic, computerized search of the literature in CINAHL (1981-present), MEDLINE (1946-present), Embase (1974-present), and EBM Reviews (1991-present) was conducted in June 2014. The reference lists of all selected articles were checked to retrieve relevant articles that may have been missed during the computerized search. The reference lists were reviewed by 1 author (C.F.F.).

Search

The following key terms were used to search databases: hamstring, biceps femoris, semimembranosus, semitendinosus, thigh, upper leg, posterior thigh, injury, leg injury, athletic injury, sport injury, strain, sprain, tear, recovery of function, recover, return to play, and sports medicine. No limit was set. See Appendix 1 (available online at http://ajsm.sagepub.com/supplemental) for the full electronic search strategies.

Study Selection

To identify relevant articles, titles and abstracts were independently screened by 2 authors (C.F.F. and P.G.). Full-text articles were read if eligibility could not be established based on the information in the abstracts. Disagreements between the reviewers were resolved by consensus.

Data Collection Process

We developed a data extraction sheet based on the Cochrane Handbook's checklist of items for data extraction.[12] One author extracted the data from the included studies, and a second author validated the extracted data once completed (C.F.F. and D.H.G.).

Data

The extracted information included methods, trial participant characteristics, type of intervention, type of outcome measure, and results.

Risk of Bias in Individual Studies

Because of the heterogeneity of the studies, we decided to assess the quality of the studies by determining their level of evidence according to the Oxford Centre for Evidence-Based Medicine's levels of evidence.[18]

RESULTS

Study Selection

The systematic search of CINAHL, MEDLINE, Embase, and EBM Reviews provided a total of 1003 citations. Eleven additional articles were identified via a manual search of the reference lists of all selected articles. Once duplicates were removed, 914 titles were retained. Furthermore, 873 articles, screened by titles and abstracts, were removed because they failed to meet the inclusion criteria. The full text of the remaining 41 articles was assessed for eligibility, and 24 studies were included in the review (Figure 1).

Study Characteristics

The sample sizes of the studies ranged between 14[21] and 397[17] athletes. Larger sample sizes came from the studies of Ekstrand et al[9] and Mohamad Shariff et al,[17] which included 393 thigh injuries, of which 298 were posterior thigh

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injuries, and 397 muscle injuries, of which 125 were hamstring injuries, respectively. These studies included male and female athletes aged 14 to 53 years. Various sports were represented, but most studies involved track and field,[1,2,13,15-17] dance,[1,3] football,[6,13,22] Australian rules football,[7,8,11,23,26-28] rugby,[7] and soccer.[5,9,10,20] Athletes ranged in level from recreational to elite international. Diagnosis criteria for hamstring injuries were not standardized. Some authors used imaging techniques,[6,7,21,22,26] while others employed a clinical assessment[13,16,24,28] or both.[1-5,8-11,17,20,23,25,27] Injuries were graded differently; however, most authors used the Peetrons classification (grade 0-III).[19] Study samples included athletes with[6,8,13,20,23,28] and without[1-4,26] previous hamstring injuries. Recovery time was expressed in terms of return to the preinjury level,[1-3] return to play,[8,20] return to sports,[4,13,17,24,25] return to full participation,[5] return to full training,[9,10] return to competition,[7,23,26-28] duration of rehabilitation[11,15,16] or convalescence,[21] days to recover,[22] or number of practices and/or games missed[6] (see Appendix 2, available online).

Risk of Bias Within Studies

Four studies were classified as having level 2 evidence,[5,16,24,25] 1 study as level 3,[15] and 19 studies as level 4[#] (Table 1).

Results of Individual Studies

All results are summarized in Table 2. The studies were organized into 3 categories: positive prognostic factors, negative prognostic factors, and factors having no effect on recovery (ie, factors that failed to achieve a statistically significant positive or negative association with outcome measures).

Synthesis of Results

Clinical Assessment. Sixteen studies reported clinical assessment factors as predictors of recovery.** Sprinting-type injuries had a better prognosis than stretching-type injuries.[5] Age,[6,13] individual or team sport,[4] side injured,[10,13] clinically assessed as a grade I or II injury,[13] straight leg raise (2-3 days after injury),[1,28] knee flexion strength (2 days after injury),[1] time taken to ascend stairs pain free,[28] slump test (within 3 days of injury),[28] active knee extension test (within 3 days of injury),[28] pain provocation test (within 3 days of injury),[28] palpated length of the painful area,[2,3] and use of nonsteroidal anti-inflammatory drugs[28] were found to have no effect on prognosis.

Factors related to longer injury recovery times were recreational-level sports,[4] a structural injury (acute, indirect muscle disorder with macroscopic evidence of muscle fiber damage) compared with functional injury (painful muscle disorder without macroscopic evidence of muscle fiber damage),[9] greater range of motion (ROM) deficit,[15,23] time to first consultation >1 week,[17] increasing pain on the visual analog scale (VAS),[23,27] and >1 day to be able to walk pain free after the injury.[28]

[#]References 1-4, 6-11, 13, 17, 20-23, 26-28.
**References 1-6, 9-11, 13, 15, 17, 22, 23, 27, 28.

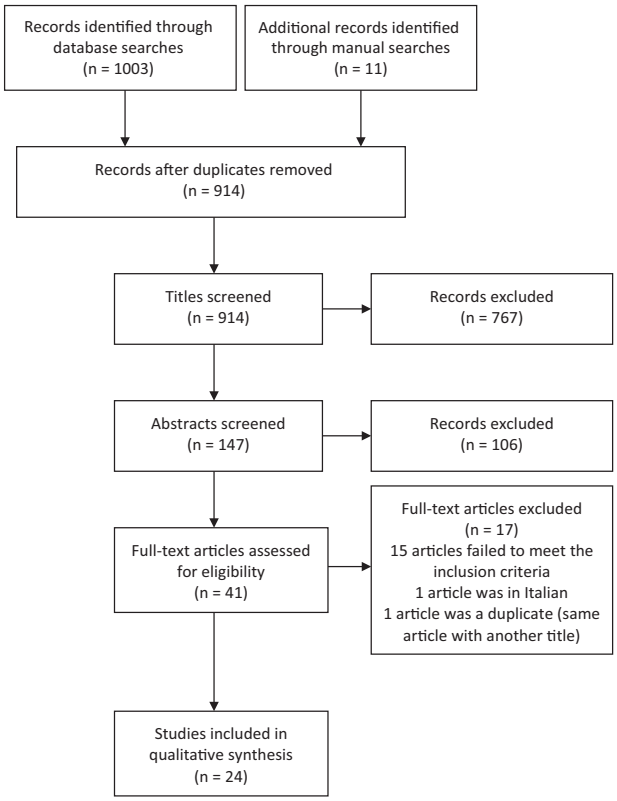

Figure 1. Study flowchart.

Some results were contradictory between studies. In 2 studies by Askling et al,[2,5] a more cranial point of pain identified with palpation was associated with a negative prognosis; however, this was not the case in 2 other studies by the same authors.[3,4] The study sample in the latter 2 studies was composed of stretching-type hamstring injuries, which are already thought to have a poor prognosis. Kilcoyne et al[13] found no association with the type of sport, whereas Askling et al[1] found that dance was a negative prognosis factor. The sample size in the study by Kilcoyne et al[13] was 48 athletes from multiple sports involving high-speed running, whereas the sample in the study by Askling et al[1] included 18 sprinters and 15 dancers. Sex had no effect in the Kilcoyne et al[13] study, but female sex had a negative effect in the study by Mohamad Shariff et al.[17] The latter studies included 360 athletes with various muscle injuries, and female sex was associated with a worse prognosis in the whole sample but not specifically with hamstring injuries. Previous hamstring injuries had no correlation with prognosis in 2 studies[11,13] but had a negative effect on recovery time in 3 studies.[17,22,28]

Imaging. Nineteen studies explored the prognosis value of imaging techniques.[††] Superficial muscle injuries,[21] small cross-sectional areas,[21] injuries not involving the proximal tendon (PT),[5] and negative magnetic resonance imaging (MRI) findings[5,8,11] were associated with a better prognosis. Positive MRI findings,[9,27] higher grade of injury

[††]References 2-11, 13, 15, 20-23, 25-27.

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TABLE 1
Levels of Evidence

Article	Return to Play: Prognosis Factors			
	Clinical Assessment or Rehabilitation	Imaging	Both	Level of Evidence
Askling et al[1] (2006)	x			4
Askling et al[2] (2007) [high speed]			x	4
Askling et al[3] (2007) [slow speed]			x	4
Askling et al[4] (2008)			x	4
Askling et al[5] (2013)			x	2
Cohen et al[6] (2011)		x		4
Comin et al[7] (2013)		x		4
Connell et al[8] (2004)		x		4
Ekstrand et al[10] (2012)			x	4
Ekstrand et al[9] (2013)			x	4
Gibbs et al[11] (2004)		x		4
Kilcoyne et al[13] (2011)	x			4
Malliaropoulos et al[16] (2004)	x			2
Malliaropoulos et al[15] (2010)			x	3
Mohamad Shariff et al[17] (2013)	x			4
Petersen et al[20] (2014)		x		4
Pomeranz and Heidt[21] (1993)		x		4
Rettig et al[22] (2008)		x		4
Schneider-Kolsky et al[23] (2006)			x	4
Sherry and Best[24] (2004)	x			2
Silder et al[25] (2013)			x	2
Slavotinek et al[26] (2002)		x		4
Verrall et al[27] (2003)			x	4
Warren et al[28] (2010)	x			4

on MRI,[6,9,10,22] muscle involvement >75%,[6] complete transection,[21] retraction,[6] central tendon disruption of the biceps femoris,[7] and PT involvement on MRI[2] were poor prognosis signs. Shorter distance to the ischial tuberosity on MRI[2,5]; length of the hamstring injury on MRI[2,5,6,8,11,23,25]; and depth,[2] volume,[2] and large cross-sectional area[2,8,11,15,21,23,26] were found to be associated with a poor prognosis in many studies but not in the 2 studies by Askling et al[3,4] involving stretching-type hamstring injuries. Pomeranz and Heidt[21] revealed that ganglion-like fluid collection and hematomas on MRI were associated with a worse prognosis, but these findings were not confirmed by Slavotinek et al.[26] It should be noted that no significant threshold was mentioned in the Pomeranz and Heidt[21] study.

Although 2 studies[8,23] found an association between an injury to the biceps femoris and a longer recovery time, 4 studies[7,10,13,26] did not find any such relationship. As for the number of muscles involved, results were also contradictory. Cohen et al[6] found that multiple muscle/tendon involvement was a worse prognosis factor. However, Gibbs et al[11] found no distinct effect between single and double muscle involvement. Lastly, some authors discovered that injuries outside the musculotendinous junction[8] and distal myotendinous junction tears[21] were associated with a poor prognosis, while others noted that the location of the injury[6,26] had no effect.

We decided to discuss the ultrasound (US) results separately because of the different features of this imaging technique compared with MRI. Negative US findings were a better prognosis determinant in the Connell et al[8] study, but no difference in prognosis was found between positive and negative US results by Petersen et al.[20] The following were found to be negative prognosis factors: cross-sectional area,[8,15] injury outside of the musculotendinous junction,[8] hematoma,[8,15] structural injury,[9] and injury involving the biceps femoris.[8] Connell et al[8] determined that a longer injury on US was a poor prognosis factor, whereas Petersen et al[20] did not observe this association. No correlation was revealed between the injured area and time to recovery.[15]

Rehabilitation. Five studies addressed rehabilitation and time to recovery.[5,16,17,24,25] Askling et al[5] showed that loading the hamstrings during extensive lengthening (L-protocol) leads to a shorter rehabilitation period compared with conventional hamstring exercises with less emphasis on lengthening (C-protocol). In addition, 4 daily static hamstring stretching sessions were better than 1 daily session in terms of rehabilitation duration.[16]

No difference was found between a static stretching, isolated progressive hamstring resistance exercise rehabilitation program and a progressive agility and trunk stabilization exercise rehabilitation program[24]; no difference was revealed between a progressive agility and trunk stabilization program and a progressive running and eccentric strengthening rehabilitation program[25] or between the frequency of physical therapy sessions.[17]

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TABLE 2
Summary of Results[a]

Positive Prognosis Factors	Negative Prognosis Factors
L-protocol vs C-protocol[5]	Dancers[1]
Sprinting-type injury vs stretching-type injury[5]	Point of greatest pain during palpation: more cranial at 2 and 21 days after injury[2]
Injury not involving the PT vs involving the PT (MRI within 5 days)[5]	Point of greatest pain during palpation: more cranial at 10 days after injury (tendency)[2]
No injury on MRI[5,11]	PT involvement (MRI)[2]
Negative imaging results within 3 days[8]	Distance to the ischial tuberosity (MRI) at 4, 10, 21, and 42 days after injury[2]
4 daily sessions of static hamstring stretching (group B) vs 1 daily session (group A)[16]	Proximal muscle-tendon junction and/or PT vs distal muscle-tendon junction and/or distal tendon/distal muscle belly (MRI) (proximal vs distal)[2]
Superficial muscle injuries[21]	Shorter distance from the most proximal pole of the injury to the ischial tuberosity measured by MRI (within 5 days) or peak pain during palpation[5]
Small cross-sectional area[21]	Length (MRI) at 21 and 42 days[2]
No Effect	Longer duration of edema (MRI within 5 days)[5]
Hip ROM (SLR) at 2 days after injury[1]	Long T2 sagittal-plane signal (MRI within 3 days)[6]
Passive SLR test (within 3 days after injury)[28]	Length of the injury (US and MRI)[8]
Knee flexion strength at 2 days after injury[1]	Length (MRI within 24-72 hours)[11]
Palpated length of the painful area[2]	Longitudinal length of the injury (MRI within 3 days)[23]
Palpated length of the painful area at 2, 10, 21, and 42 days[3]	Initial craniocaudal length of the injury (MRI within 5-7 days of injury)[25]
Width (MRI) at 4, 10, 21, and 42 days[2]	Depth (MRI) at 4, 10, 21, and 42 days[2]
Point of greatest pain during palpation: more cranial at 2, 10, 21, and 42 days[3]	Volume (MRI) at 4, 10, 21, and 42 days[2]
Location of most pain during palpation[4]	Cross-sectional area (MRI) at 4, 10, and 21 days[2]
Region, length, width, depth, volume, and cross-sectional area (MRI at 2, 10, 21, and 42 days)[3]	Cross-sectional injury area (as a percentage score) (US and MRI)[8]
Length of the injury (MRI at a mean 13 weeks [range, 1-52 weeks])[4]	Cross-sectional area (MRI within 24-72 hours)[11 b]
Length of the injured area (US at 1-10 days after injury)[20]	Cross-sectional area >25% (US at 48 hours after injury)[15]
Distance to the ischial tuberosity (MRI at a mean 13 weeks [range, 1-52 weeks])[4]	Cross-sectional involvement >50%[21 b]
Individual vs team sports[4]	Cross-sectional area[23]
Location of the hamstring injury (MRI within 3 days)[6]	Cross-sectional area >50% (MRI within 3 days)[26]
Injured area[15,26-28]	Recreational level vs elite level[4]
Age[6,13]	Grade III vs grades I and II (MRI within 3 days)[6]
Muscle involved (MRI)[7,10,13,26]	Higher grade on MRI (0-III) (within 24-48 hours)[10]
Injury to the kicking leg vs supporting leg[10]	Subgroups of structural injuries (higher): minor partial tear, moderate partial tear, subtotal/complete muscle injury/tendinous avulsion[9]
Side (right, left)[13]	Higher grade on MRI (no time frame specified): grade I, edema within the muscle belly measuring <8 cm; grade II, edema >8 cm with a vertical split in fascial planes; and grade III, tendon separation at the musculotendinous junction[22 b]
Single vs double muscle involvement (MRI within 24-72 hours)[11]	Multiple muscle-tendon involvement (>1) (MRI within 3 days)[6]
Previous hamstring injury[11,13]	High percentage of muscle involvement (>75%) (MRI within 3 days)[6]
Sex[13]	Retraction (MRI within 3 days)[6]
Sport[13]	Central tendon disruption of the biceps femoris (MRI)[7]
Degree of injury (I, II [clinically graded])[13]	Presence of an injury in the biceps femoris (US and MRI)[8]
Frequency of physical therapy sessions[17]	Site of the injury: biceps femoris[23]
Positive vs negative US findings[20]	Injury outside of the musculotendinous junction (US and MRI)[8]
Static stretching, isolated progressive hamstring resistance vs progressive agility and trunk stabilization exercise rehabilitation program[24]	Intermuscular hematoma (US)[8]
Progressive agility and trunk stabilization vs progressive running and eccentric strengthening rehabilitation program[25]	Presence of hematoma (US at 48 hours)[15]
Injury involving both the muscle belly and musculotendinous junction (MRI within 3 days)[26]	Hemorrhage-like signal intensity[21 b]
Ganglion-like fluid collection (MRI within 3 days)[26]	Presence of focal intramuscular T2 hyperintensity (MRI within 3 days)[26]
Hemorrhage-like signal intensity (MRI within 3 days)[26]	Structural posterior thigh injuries vs functional posterior thigh injuries (clinical examination, MRI, or US)[9]
Time taken to ascend stairs without pain[28]	Higher "active ROM deficit" at 48 hours after injury (classification: I, <10°; II, 10°-19°; III, 20°-29°; IV, >30°)[15]
Slump test (within 3 days after injury)[28]	Time to first consultation >1 week[17]
Active knee extension test (within 3 days after injury)[28]	Recurrent muscle injury[17,22c,28]
Use of nonsteroidal anti-inflammatory drugs[28]	Female sex[17]
Mechanism of injury (running or not)[28]	Complete transection (MRI, with no time frame specified)[21 b]
Pain provocation test (within 3 days after injury)[28]	Ganglion-like fluid collection[21 b]
	Distal myotendinous junction tears[21 b]
	Grade of injury assessed during a clinical test (pain, deficit in ROM) (within 3 days)[23]
	Positive findings on MRI (2-6 days after injury)[27]
	Increasing pain (visual analog scale)[27]
	>1 day to be able to walk pain free after the injury[28]

[a]MRI, magnetic resonance imaging; PT, proximal tendon; ROM, range of motion; SLR, straight leg raise; US, ultrasound.
[b]Significance not mentioned.
[c]Tendency value.

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DISCUSSION

Summary of Evidence

Our study investigated the determinants that best predict return to play after a hamstring injury. We identified numerous studies focusing on this subject. Unfortunately, the strength of the evidence varied greatly across these studies, and some of the results were contradictory. Nevertheless, a summary of the key factors needing consideration, modulated according to the strength of the evidence, was possible. With regard to the clinical assessment, stretching-type injuries,[5] recreational-level sports,[4] structural versus functional injuries,[9] greater ROM deficit,[15,23] time to first consultation >1 week,[17] increased maximal pain score on the VAS,[23,27] and >1 day to be able to walk pain free after the injury[28] were associated with longer recovery times. As for MRI, the following correlated with longer recovery times: studies demonstrating a hamstring injury (positive findings)[9,27]; higher grade of injury[6,9,10,22]; >75% muscle involvement[6]; complete transection[21]; retraction[6]; central tendon disruption of the biceps femoris[7]; PT involvement[2]; shorter distance to the ischial tuberosity[2,5]; length of the injury[2,5,6,8,11,23,25]; and depth,[2] volume,[2] and large cross-sectional area.[2,8,11,15,21,23,26] In terms of US, large cross-sectional area,[8,15] injury outside of the musculotendinous junction,[8] hematoma,[8,15] structural injury,[9] and injury involving the biceps femoris[8] were found to have a negative prognosis. Lastly, rehabilitation that incorporated hamstring loading during extensive lengthening[5] or 4 daily sessions of static hamstring stretching[16] led to a shorter rehabilitation period.

Four studies met level 2 evidence criteria based on the Oxford Centre for Evidence-Based Medicine's levels of evidence.[18] Askling et al[5] demonstrated that a sprinting-type injury, an injury not involving the PT, an MRI-negative injury, and the use of the L-protocol (including hamstring loading during extensive lengthening) in rehabilitation accelerated return to play. In this study, a shorter distance to the ischial tuberosity and a longer duration of edemas seen on MRI prolonged the time needed before returning to play. The 3 other studies focused on rehabilitation programs. Malliaropoulos et al[16] found that 4 daily sessions of static hamstring stretching instead of 1 daily session reduced recovery time. Sherry and Best[24] and Silder et al[25] did not find any statistical differences in terms of time needed before returning to play between a static stretching, isolated progressive hamstring resistance exercise rehabilitation program and a progressive agility and trunk stabilization exercise rehabilitation program and between a progressive agility and trunk stabilization program and a progressive running and eccentric strengthening rehabilitation program, respectively.

Limitations

One limitation of the study relates to the instrument used to assess the level of evidence. Because of the number of studies analyzed and the variable quality of these studies, the use of the Oxford Centre for Evidence-Based Medicine's levels of evidence[18] seemed to be the best instrument for our review. Only 4 of the studies analyzed were randomized trials. It was difficult to draw general conclusions, given the variability of the diagnosis criteria and outcome measures in addition to the population's heterogeneity across the studies reviewed. Furthermore, we limited the review to English- or French-language articles.

CONCLUSION

Numerous determinants positively or negatively affect return to play after a hamstring injury in athletes. In clinical practice, key aspects to look for that may foreshadow a longer recovery time are stretching-type injuries,[5] recreational-level sports,[4] structural injuries (macroscopic evidence of muscle fiber damage),[9] greater than 20° to 25° of active ROM deficit of the knee,[15,23] time to first consultation >1 week,[17] increased maximal pain score on the VAS of ≥5,[23,27] and >1 day to be able to walk pain free after the injury.[28] Either MRI or US can provide complementary information (ie, determinants or predictors) on the biological integrity of the hamstring and surrounding tissues that helps predict return to play. Finally, rehabilitation that incorporated hamstring loading during extensive lengthening[5] (ie, eccentric training protocol) or 4 daily sessions of static hamstring stretching[16] led to shorter rehabilitation times. It is important for sports professionals to be aware of these factors to guide athletes' rehabilitation and return to play. Additional high-quality research studies are needed to reinforce the strength of the evidence for these determinants to improve the clinical decision-making process.

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